Document ID: BSC-SEC-2026-V3
Date: November 2026
Classification: Public Reference Document

Smart Contract Internal Security & Architecture Assessment

White-box code evaluation, invariant verification and static analysis of the **BSCook Protocol** Single-Sided Uniswap V4 Liquidity Engine on BNB Smart Chain.

0	0	CONFIRMED
CRITICAL VULNERABILITIES	HIGH-SEVERITY FINDINGS	NO DIRECT EXPLOITS IDENTIFIED

1. Technical Summary

This document presents the internal technical and security review conducted by the **BSCook Core Engineering Team** across the protocol's 6 Solidity smart contracts (3,553 lines of code). The evaluation examined potential attack vectors, economic invariants, reentrancy vulnerabilities, access control bounds, and liquidity locking mechanisms.

Core Finding: NO DIRECT FUND-LOSS EXPLOITS IDENTIFIED. Locked LP positions cannot be withdrawn by either users or administrators. Tokens launched through the factory are standard OpenZeppelin ERC20 clones with zero post-initialization minting, zero transfer taxes, and zero honeypot capabilities.

2. Protocol Specifications

Protocol Name	BSCook Protocol	**Deployment Network**	BNB Smart Chain (BSC Mainnet - Chain ID: 56)
Official Domain	https://bscook.fun	**DEX Architecture**	Single-Sided Uniswap V4 Pool Initialization
Token Supply Model	1,000,000,000 Fixed Supply (18 Decimals)	**LP Vault**	BSCookLocker (Non-Custodial, Permanent LP Lock)
Solidity Compiler	v0.8.20 (viaIR: true, runs: 800)	**Dev-Buy Cap**	0.32 BNB hard-cap per creator launch transaction

3. Scope of Assessed Contracts

Contract Name	Mainnet Contract Address	Source Code Status
BSCookFactory	`0xe8a0beEEdd39af3a87De11d6b120D0fFa874867D`	**Verified on BscScan**

Contract Name	Mainnet Contract Address	Source Code Status
BSCookToken (Impl)	`0x66aEfEE3B1f3163f036Aeee733035df09F49Fc02`	**Verified on BscScan**
BSCookLocker	`0x3a8587057f85eb318F52c2C9ae148940741CFCfD`	**Verified on BscScan**
BSCookTimelock	`0x4cec2C1DA4D2f460402aC38aF04Ed29D0C922c83`	**Verified on BscScan**
BSCookMultiSig	`0xAa2699E1737052869c97a681d3d6d1C9b969120e`	**Verified on BscScan**
ContractValidation	Internal Shared Library	**Verified on BscScan**

4. Core Security Invariants Verified

Security Invariant	Code Verification Details	Status
Rug-Pull Immunity	`BSCookLocker` possesses NO functions to reduce or withdraw locked LP liquidity. `rescueERC721` explicitly forbids interacting with locked position IDs.	CONFIRMED IMMUTABLE
Honeypot & Blacklist Immunity	`BSCookToken` is a clean OpenZeppelin ERC20 implementation without transfer fees, blacklist arrays, trading pauses on users, or post-init owner privileges.	CONFIRMED IMMUTABLE
Admin Asset Protection	In `BSCookFactory`, `rescueERC20` explicitly protects user tokens and WBNB from arbitrary admin withdrawal.	CONFIRMED RESTRICTED
Single-Sided Math Correctness	TickMath constants, initial prices (`INITIAL_WBNB_QUOTE = 6 ether`), tick ranges (`MAX_USABLE_TICK = 887200`), and liquidity math independently verified for correctness.	VERIFIED SOUND
Anti-Sniper Dev-Buy	Initial creator purchase is hard-capped on-chain at `0.32 BNB`, preventing creator monopoly of supply at genesis.	VERIFIED ENFORCED
Reentrancy Defense	External mutating routines protected with OpenZeppelin `ReentrancyGuard`. CEI strictly followed.	VERIFIED SOUND

5. Governance & Centralization Transparency

Mechanism	Code Implementation	Risk & Security Consideration
Future Locker Migration	`queueLockerChange` allows owner to propose a new locker address with a mandatory **2-day queue delay**.	CENTRALIZATION RISK Existing launches remain permanently safe in the original locker; however, future launches after the 2-day delay deposit LP to the new locker. Community must monitor `LockerChangeQueued` event.
Creator Fee Freezing	`emergencyFreezeWallet` / `emergencyFreezeFees`: Fee Guardian (2-of-3 MultiSig) can freeze up to **7 days**. Timelocked owner can freeze **indefinitely**.	TRUST IN GOVERNANCE Creator's 75% fee share can be frozen if reported compromised. Only Timelocked Owner can unfreeze after verification. Underlying LP is never affected.
MultiSig & Timelock	Critical operations execute through `BSCookTimelock` (12h–48h delay) and `BSCookMultiSig` (2-of-3 signers).	TIME-DELAYED DEFENSE Requires multi-signature consensus and provides 12 to 48 hours advance warning of queued proposals.

6. Static Analysis Evaluation (Slither)

Static analysis was executed across the codebase using **Slither**. Automated reports flagged 6 high-level alerts (weak-prng, reentrancy-balance); manual code review determined these to be false positives resulting from tick modulo calculations and nonReentrant Uniswap calls.

7. Disclaimer

This document represents an internal engineering and security assessment conducted by the BSCook Core Team for architecture transparency. It does not replace an accredited commercial third-party security audit. Interacting with decentralized smart contracts carries market and smart contract risks.